4.3.1 Modeling with effective medium theories

This section is devoted to better investigate the influence of the dielectric media on the plasmonic properties of Au NPs in sandwiches and to discern among competing events. The choice has been driven towards sandwiches because the effect of permittivity change in bottoms was too weak to identify a clear and reproducible trend in LSPR shift, thus theoretical models have been applied only to fully embedded particles. Optical responses (LSPR position) have been modeled within the framework of the EMT approach [116, 118, 306], able to well-suite the dielectric properties of inhomogeneous mixtures of NPs buried in a media [9]. However, the accuracy of this analysis is limited by numerous assumptions and in this work should be intended only as a qualitative tool to roughly comprehend LSPR trend with the surrounding and not for the precise prediction of absolute values.

Practically, the method consists in replacing the particle permittivity (for gold ϵ_{Au}) with an effective one (ϵ_{eff}), characteristic of the whole compound, obtained by averaging with ad-hoc mixing rules for the dielectric functions of host medium (ϵ_h) and inclusions [41, 42, 404]. This approach is crucial in order to get a correct reproduction of the *effective* optical output that widely surpasses responses of individual components. Indeed, as a consequence of the previous hypothesis, the overall mixture exhibits a uniform polarizability, meaning that the electromagnetic field is static [405] and interacts with an homogeneous mixture on average. Then, a prediction of the LSPR wavelength has been attempted by substituting ϵ_h with ϵ_{eff} in the conventional Mie resonance for spherical NPs [1, 42, 406]. The ϵ_{Au} used in all the computations have been extracted from the linear approximation [406] of experimental data for thin gold films (taken from Johnson e Christy [15]) calculated at 700 nm. Only the real part has been considered, as it prevails on the complex component leading to $\epsilon_{Au}(\omega) = \epsilon_{1,Au}(\omega)$:

$$\epsilon_{1,Au} = -0.0744\lambda + 35.22 \tag{4.1}$$

The theory holds under many hypotheses, to begin with extension of nanostructures kept well below the wavelength of incident radiation, i.e. the quasistatic regime ($D << \lambda$) [404], and that the particles can be well recognised as sporadic inclusions in a prevailing host material. As a first estimate, the well-known Maxwell-Garnett (MG) mixing formula [306] has been used to deduce the effective permittivity $\epsilon_{eff} = \epsilon_{MG}$ of the film:

$$\frac{\epsilon_{MG}(\omega) - \epsilon_h}{\epsilon_{MG}(\omega) + 2\epsilon_h} = f\frac{\epsilon_{Au}(\omega) - \epsilon_h}{\epsilon_{Au}(\omega) + 2\epsilon_h} \tag{4.2}$$

where $\epsilon_{Au}(\omega)$ and ϵ_h are the dielectric functions (in principle complex) of Au NPs and host respectively. This model is very powerful because of its intrinsic versatility to be applied in principle to any type of composite media, just by inserting the proper values of the dielectric constants. The parameter f is the volume filling factor of inclusions, which refers to the relative gold amount taken as the ratio between the nominal evaporated thickness (3 nm) over the total composite thickness (70 nm) [406]. Besides, ϵ_h has been evaluated from ellipsometry on Au-free TCO samples with similar thickness (Figure SI4), by taking the average value of the real part $\epsilon_{1,h}$ in the wavelength range required (1.5-2.5 eV, i.e. 550-800 nm). In this window of energies, the imaginary permittivity $\epsilon_{2,h}$ is relatively lower than $\epsilon_{1,h}$, i.e. practically zero for TiO_2 and below 1 for Ta:TiO_2 films. These data support the use for $\epsilon_{1,h}$ in the computations and the matrix can be modeled as a non-adsorbing media [15, 406]. Hence, the LSPR position has been calculated by inserting ϵ_{MG} in the conventional Frölich condition [1, 389, 406, 407]:

$$\lambda_{LSPR} = \lambda_p\sqrt{1 + 2\epsilon_{MG}(\omega)} \tag{4.3}$$

Here, λ_p is the bulk plasma wavelength of gold, estimated at 145 nm [9]. It is worth noting that equation 4.1 can be analytically solved just by knowing dielectric permittivities and f, thus it

is applicable in principle to any metal-dielectric pair. MG is valid only for diluted media ($f << 1$) in which NPs should be spherical, homogeneously distributed and isolated within the volume [9, 408]. Despite the versatility of this approach, MG model provides an effective function lacking in size or shape dependency. The absence of any geometrical contribution is still a valid approximation as long as the nanocomposite is uniform and isotropic on average [408]. Nevertheless, the high degree of randomness and intrinsic anisotropy of the systems restrict its effectiveness.

To overcome these limitations, an improved Maxwell-Garnett approach has been proposed, called Dynamic Maxwell-Garnett (DMG) theory [406, 409], that accounts for *(i)* real NP shapes (oblate spheroids-ellipsoids), as a deviation from the classical sphere of the Mie theory is expected *(ii)* real eccentricity of NPs modeled as ellipsoids (length/height acquired from SEM images). The simplified version of DMG for a two-phase composite provides the effective permittivity ϵ_{DMG}:

$$\epsilon_{DMG} = \epsilon_h \frac{\epsilon_{Au} + K\epsilon_h + fK(\epsilon_{Au} - \epsilon_h)}{\epsilon_{Au} + K\epsilon_h - f(\epsilon_{Au} - \epsilon_h)} \tag{4.4}$$

Where the geometry dependency is introduced through the shape factor Ki:

$$K_i = \frac{1}{L_i} - 1 \tag{4.5}$$

The depolarisation factor L_i expresses the oscillation of the electromagnetic field along the principal axes of the ellipsoid, implying $\sum_{i=1}^{3} L_i = 1$ [1, 406, 409]. In the case of spherical inclusions, $L_x = L_y = L_z = 1/3$, otherwise $0 < L_i < 1$. For simplicity, only polarization of the horizontal in-plane direction has been assumed hence, for oblate-shaped NPs, $L_x = L_y$. The ratio between NP diameter and height is directly embodied in the expression of L_i through the ellipticity (diameter/height) [406, 407, 409]. As a consequence of the Frölich condition of resonance [406], the LSPR wavelength has been evaluated straightforward from:

$$\lambda_{LSPR} = 473 + 13.4K\epsilon_h + 13.4(K + 1)\epsilon_h \tag{4.6}$$

However two considerations should be made on the choice of the real part $\epsilon_{1,Au}$ for both MG and DMG models [410]. First, interband transitions should be accounted for, as gold is subject to strong electronic excitations from the d band over the Fermi energy at the threshold of 2.4 eV (516 nm). Weaker excitation pathways from $5d$ to $6sp$ bands are also expected approaching 1.8 eV (688 nm), at which corrections should be introduced accordingly. Second, size-confinement effects become important when dimensions fall below 30 nm, along with higher probability of defect states and impurities at interfaces [60]. Retardations in electron oscillation may arise when the NP size is comparable to the mean free path of electrons (for gold around 42 nm) [410] thus surface scattering mechanisms prevail, justifying the need for a size-dependent dielectric function with additional dephasing terms. However, size effects are significant only below 10 nm [411, 412]. As previously discussed, at LSPR wavelengths (550-800 nm) Au interband excitations are weak, hence employment of the real part only $\epsilon_{1,Au}$ (calculated at 700 nm with Eq. 4.1) is acceptable within the aim of a qualitative description of the plasmon shift as a function of the environment. The geometrical parameters and $\epsilon_{1,h}$ used in MG and DMG models have been summarised in Table 4.1. It should be noted that f, total thickness and height of NPs were kept fixed for all systems, in order to avoid excessive variability responsible for an increase in the complexity of results. Anyhow this choice is reasonable as morphological features are almost unaffected when changing the type of matrix.

Matrix type	Average ϵ_{1m} (550-800 nm)	Thickness (nm)	Height (nm)	f	DMG parametersDMG parameters	
					Height (nm)	D_{mean} (nm)
TiO_2	6.2		16.5			
Ta(5%):TiO_2	5.6	70	13	0.042	15	25.5
Ta(10%):TiO_2	5.4		14.7			

Table 4.1: *Summary of Au NPs geometrical parameters characterizing Au-sandwiches in different matrices, with specific parameters used in DMG model.*

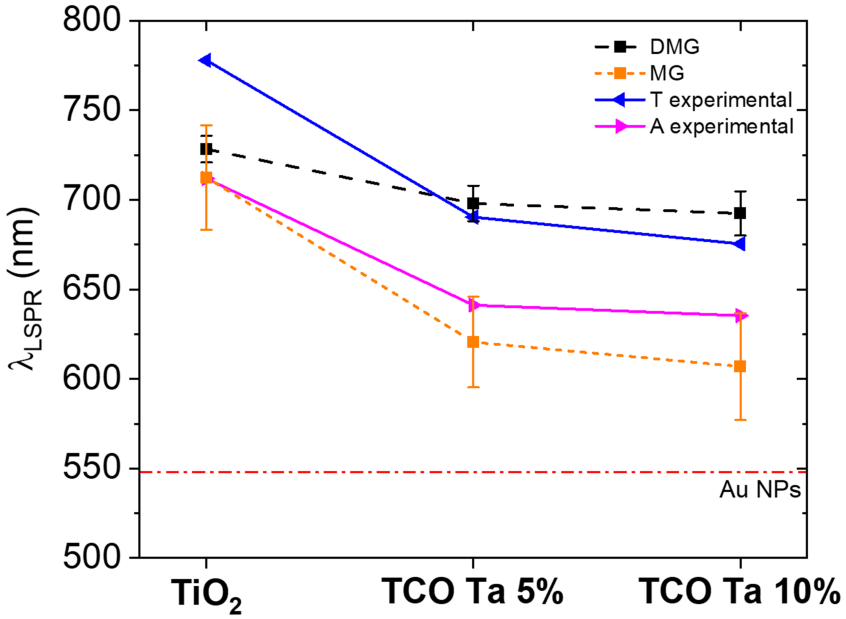

Figure 4.11: *λ_{LSPR} predicted from MG and DMG models as a function of the matrix type, for Au-sandwiches. Experimental λ_{LSPR} obtained from direct transmittance and absorbance curves are shown for comparison.*

The output LSPR positions given by the models are displayed in Fig. 4.11, compared with experimental resonances fitted from transmittance or absorbance spectra. MG model reflects the experimental trend in λ_{LSPR} extracted form absorbance, in excellent agreement for TiO_2 sandwich, although predicted values deviate from experimental findings at higher Ta contents. However, the good matching of theoretical trends to experiments proves that the main effect causing λ_{LSPR} to blue-shift is the reduction of $\epsilon_{1,h}$.

Finally, the DMG model has been applied keeping fixed the average diameter D_{mean} at 25.5 nm for the three systems. The blue-shift trend of the resonance for lower permittivities in case of DMG is weak compared to MG results. In particular λ_{LSPR} passes from 728 nm in TiO_2 sandwich to roughly 698 nm in both TCO composites. In addition, there is a good agreement with LSPR positions extracted from transmission spectra. This behaviour is representative of the role of the shape factor K, related to the ellipticity of NPs. In fact, the introduction of geometrical dependency into EM models results in an overall loss in reliability and the effect of the dielectric function of the matrix is overwhelmed by some extra contributions, not easy to discern. Its applicability is restricted by the several approximations initially stated, making it inaccurate for a precise description of complex optical phenomena arising in the films. In fact when trying to compute the filling factor by reversing equation 4.6 with experimental λ_{LSPR}, f loses physical meaning becoming negative or diverging (20%) in comparison with the one calculated according to experimental considerations (4.28%). This discrepancy is inherent in the assumption that the whole TCO participates in forming the effective medium, although

[1] Stefan A. Maier. Plasmonics: Fundamentals and Applications. Springer US, New York, NY, 2007.

[116] Tuck C. Choy. Effective medium theory: principles and applications. Number 165 in International series of monographs on physics. Oxford University Press, Oxford, second edition edition, 2016. OCLC: ocn938790981.

[118] G. A. Niklasson, C. G. Granqvist, and O. Hunderi. Effective medium models for the optical properties of inhomogeneous materials. Appl. Opt., 20(1):26–30, Jan 1981.

[306] ARI SIHVOLA. Electromagnetic Metamaterials: Homogenization and Effective Properties of Mixtures, chapter CHAPTER 1, pages 1–36. 2019.

[9] Sergey V. Gaponenko. Introduction to Nanophotonics. Cambridge University Press, 1 edition, April 2010.

[41] Vincenzo Amendola, Roberto Pilot, Marco Frasconi, Onofrio M Marag., and Maria Antonia Iat.. Surface plasmon resonance in gold nanoparticles: a review. Journal of Physics: Condensed Matter, 29(20):203002, apr 2017.
[42] MA Garcia. Surface plasmons in metallic nanoparticles: fundamentals and applications. Journal of Physics D: Applied Physics, 44(28):283001, jun 2011.
[404] Localized surface plasmon resonance in au nanoparticles embedded dc sputtered zno thin films. Journal of Nanoscience and Nanotechnology, 15(2), 2015.
[406] Ranjit Laha, P. Malar, Thomas Osipowicz, and S. Kasiviswanathan. Tailoring plasmonic properties of metal nanoparticle-embedded dielectric thin films: the sandwich method of preparation. J Nanopart Res, 19(9):302, 2017.

[407] N. G. Khlebtsov, L. A. Trachuk, and A. G. Mel'nikov. The effect of the size, shape, and structure of metal nanoparticles on the dependence of their optical properties on the refractive index of a disperse medium. Opt. Spectrosc., 98(1):77–83, 2005.
[408] Vadim A. Markel. Introduction to the maxwell garnett approximation: tutorial. J. Opt. Soc. Am. A, 33(7):1244–1256, 2016.
[409] Stefano Giordano. Effective medium theory for dispersions of dielectric ellipsoids. Journal of Electrostatics, 58(1):59–76, 2003.

[389] D. Buso, J. Pacifico, A. Martucci, and P. Mulvaney. Gold-nanoparticle-doped tio2 semiconductor thin films: Optical characterization. Advanced Functional Materials, 17(3):347–354, 2007.

[60] Audrey Moores and Fr.d.ric Goettmann. The plasmon band in noble metal nanoparticles: an introduction to theory and applications. New J. Chem., 30:1121–1132, 2006.

[410] Anastasiya Derkachova, Krystyna Kolwas, and Iraida Demchenko. Dielectric Function for Gold in Plasmonics Applications: Size Dependence of Plasmon Resonance Frequencies and Damping Rates for Nanospheres. Plasmonics, 11(3):941–951, 2016.
[411] Tigran V. Shahbazyan. Landau damping of surface plasmons in metal nanostructures. Phys. Rev. B, 94:235431, 2016.
[412] Krystyna Kolwas. Decay Dynamics of Localized Surface Plasmons: Damping of Coherences and Populations of the Oscillatory Plasmon Modes. Plasmonics, 14(6):1629–1637, 2019.